MODULE: MICROBIOLOGY

Concept of single celled organisms:

Most microbes consist of a single cell. Microbes range in size from small viruses (20 nm in diameter) to large protozoans (5 mm or more in diameter). The major groups of organisms studied in microbiology, like bacteria, algae, fungi, viruses and protozoa are widely distributed in nature.

The vast majority of microorganisms are directly or indirectly beneficial, not only to other organisms, but also to humans. They form essential links in many food chains that produce plants and animals that humans eat. Aquatic microbes serve as food for small microscopic animals that in turn serve as food for fish and shellfish that humans eat. Certain microorganisms live in the digestive tracts of grazing animals such as cattle and sheep and aid in their digestive processes. Humans occasionally eat microbes, such as some algae and fungi directly.

Biochemical reactions carried out by microbes also are used by the food industry to make pickles, sauerkraut, yogurt and other dairy products. Fermentation reactions in microorganisms are used in the brewing industry to make beer and wine, and in baking to leaven dough.

The most significant benefits that microorganisms provide is their ability to synthesize antibiotics, substances derived from one microorganism that kill or restrict the growth of other microorganisms. Therefore, microorganisms can be used to cure diseases as well as cause them. Microorganisms are the major tools of genetic engineering.

Concept of species and strains:

Species are groups of actually or potentially interbreeding populations that are reproductively isolated from other such groups. Thus, under the biological species concept, species are simultaneously a reproductive community, a gene pool and a genetic system.

In biology, a strain is a genetic variant, a subtype or a culture within a biological species.

Thus it can be concluded that, where species refers to organisms that are related based on genetic, biochemical and phenotypic criteria, a strain refers to isolates of a given species based on a number of characteristics including serotyping, enzyme type, functional traits and protein plasmid characterization etc.

Identification, classification and ecological aspect of different microorganisms:

1. **Bacteria:**

   All bacteria (sing: bacterium) are single celled organisms belonging to the kingdom Monera, with spherical, rod or spiral shapes, but a few types form filaments. Most are so small that they can be seen with a light microscope only under the highest magnification. Although bacteria are cellular, they do not have a cell nucleus, and they lack the membrane-enclosed intracellular structure (organelles) found in most other cells. Many bacteria absorb nutrients from their environment, but some make their own nutrients by photosynthesis or other synthetic processes. Some are stationary, and others move about with the help of flagella. Bacteria are widely distributed in nature, they are not only found in soil and air but also in aquatic

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environments and in decaying matter. Some bacteria are pathogenic and cause diseases.

2. **Archaebacteria:**
   They are very similar to bacteria and are commonly known as archae. Archae and the bacteria belong to the same kingdom, Monera. Archae are single celled and do not have a nucleus. Many archae are extremophiles, preferring to live in environments having extreme temperatures, pH, salinities and hydrostatic and osmotic pressures. Their lipids, cell walls and flagella differ considerably from those of bacteria. Archae are not proven to cause disease in humans; in fact, many are very important in ruminant animal digestive tracts, for example, methanogens occur in gastrointestinal tract of cattle. Methanogens are responsible for production of biogas from the dung of cows, buffaloes etc.

3. **Algae:**
   Many algae (sing: alga) are single-celled microscopic organisms, but some marine algae are large, relatively complex, multicellular organisms belonging to kingdom Plantae. Algae have a clearly defined cell nucleus and numerous membrane-enclosed intracellular structures. All algae photosynthesize their own food as plants do and some can move about. Algae are widely distributed in both fresh water and oceans. Since they are so numerous and because they capture energy from sunlight and prepare their own food, algae are an important source of food for other organisms. Commercially some algae are source of iodine, bromine, agar agar, proteins etc.

4. **Fungi:**
   Many fungi (sing: fungus), such as yeasts and some molds are single-celled microscopic organisms. Some, such as mushrooms, are multicellular, macroscopic organisms. Fungi have a cell nucleus and organelles and are enveloped by chitinous cell wall. All fungi absorb ready-made nutrients from their environment. Such deriving of nutrients from other organism is called as saprophytic nutrition. Some fungi form extensive networks of branching filaments, but the organisms themselves generally do not move. Fungi are widely distributed in water and soil as decomposers of dead organisms. Some are important in medicine, for example, the first antibiotic penicillin was obtained from fungi *Penicillium sp.*

5. **Viruses:**
   Viruses are acellular entities, too small to be seen with a light microscope. They are seen with high resolution electron microscope. They are composed of specific chemical substances – a nucleic acid and a few proteins. Indeed, some viruses can be crystallized and stored in a container on a shelf for years, but they retain the ability to invade host cells. Viruses replicate themselves and display other properties of living organisms only when they have invaded host cells. They do not have their own metabolic machinery but use host cell's machinery for its reproduction thus deprive host cells from nutrition and ultimately killing them. Many viruses are known to invade human cells and cause disease.

6. **Protozoa:**
   Protozoa (sing: protozoan) also are single-celled, microscopic organisms belonging to the kingdom Protista, with at least one nucleus and numerous intracellular structures. Many protozoa obtain food by engulfing or ingesting smaller microorganisms. Most

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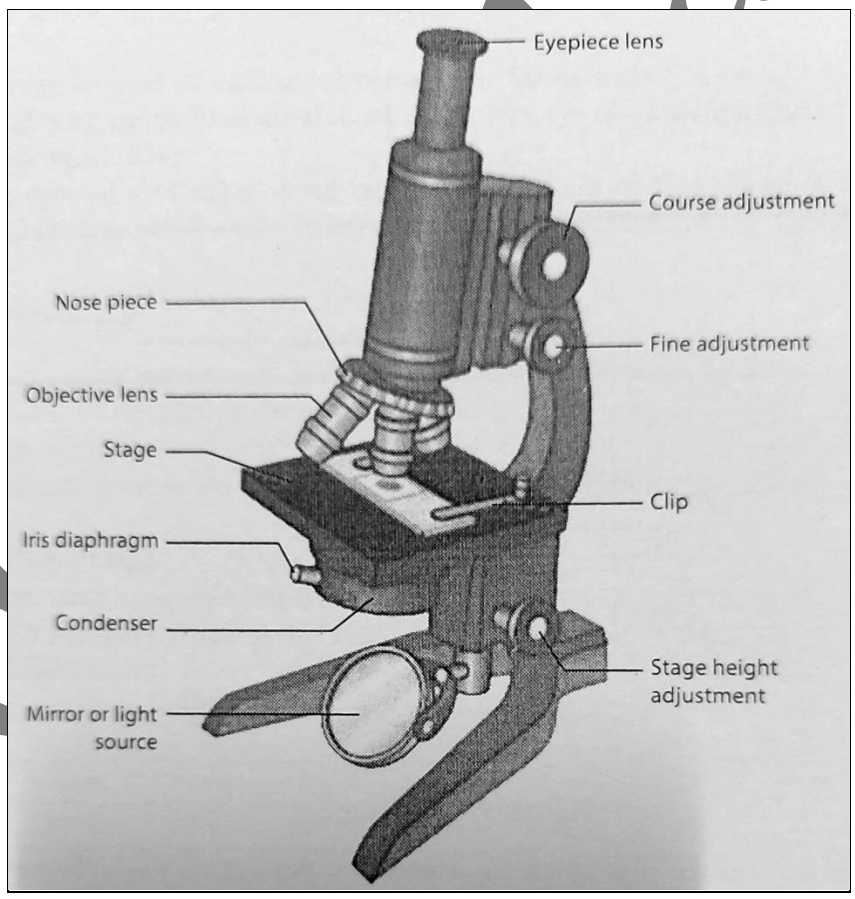
protozoa can move, but a few, especially those that cause human disease, cannot. Protozoa are found in variety of water and soil environments, as well as in animals as parasites.

Microscopy:

Introduction:
The existence of microorganisms was unknown until the invention of the microscope. Anton van Leeuwenhoek, probably the first to see microorganisms, set the stage for microscopy, the technology for making very small things visible to the human eyes.

Principle of microscopy:
Microscope is an optical instrument consisting of lens or combination of lenses for making enlarged or magnified images of minute objects. Resolution is the ability to see two objects as separate discrete entities. The resolving power of an optical instrument is its ability to separate the images of two objects, which are close together.

Light microscopy:
The major parts of a compound light microscope and their functions are as follows:

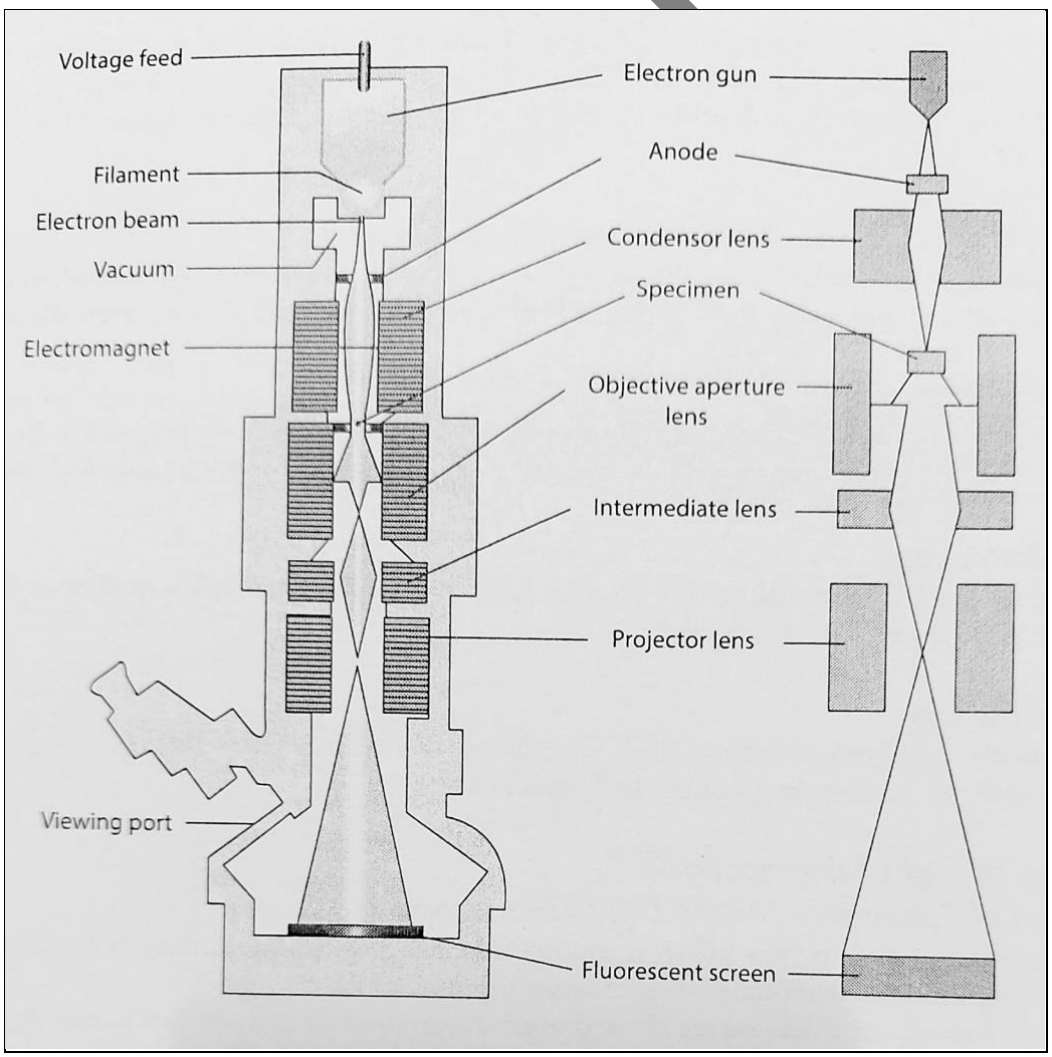


1. Mirror or light source: It may be natural visible light or inbuilt light sources such as mercury lamps, xenon lamps, LEDs, etc.
2. Condenser: Light goes through condenser which converge the light beam that passes through the specimen (transmitted light) or is reflected back from the specimen (reflected light).

3. Iris diaphragm: It can be adjusted for the amount of light reaching the specimen and into the objective lens.
4. Objective lens: It magnifies the image before it passes through body tube into the eyepiece or ocular which further magnifies the image.
5. Nose piece: It holds objective lenses of different magnification and they can be adjusted as per the requirement of observation of specimens.
6. Stage: it holds slides or specimen and the clip keeps the slide in a fixed position.
7. Coarse and fine adjustment knob: For focusing, the coarse and fine adjustment knobs are used. Coarse adjustment changes the distance between specimen and objective lens rapidly and it locates the specimen. The fine adjustment change distance very slowly and focus the specimen sharply.
8. Eyepiece: The image is either viewed directly by eye on the eyepiece or it is most often projected onto a detector, for example photographic film or more likely a digital camera.

Electron microscopy:

The electron microscope uses a beam of electrons instead of a beam of light and electromagnets instead of glass lenses for focusing. They are much more expensive, sophisticated and difficult to use but give magnifications of up to 500,000X and a resolving power of less than 1 nm. Viruses can be seen only by using electron microscopes. Advanced types of electron microscopes can visualize actual molecules and individual atoms.



In fundamental terms electron microscope is derivative of compound microscope. The different parts of an electron microscope are shown in the figure. It also has two lens systems, a condenser, specimen holder and system for focusing the object. Electron beam is produced by electron gun which moves through vacuum to ring shaped electromagnetic lens which focus electron beam on the specimen. The electron must travel through a vacuum to avoid collision of electron beam with air molecules as it would scatter the electron and distort the image. Ultra-thin section of dehydrated specimen treated with dye is placed on metal grid in place of glass slide. Image formed by electron beam is made visible as light image on a florescent screen or monitor.
There are two types of electron microscope, Transmission electron microscope (TEM), which is typically used to examine internal cell structure and the other Scanning electron microscope (SEM), which is used to create image of the surface of the specimen and that can be examined in detail.

Sterilization:

Sterilization is the killing or removal of all microorganisms in a material or on an object. When properly carried out, sterilization procedures ensure that even highly resistant bacterial endospores and fungal spores are killed. The sterilizing agents can be broadly classified into two type, they are:

1. **Chemical agents:**
   They kill microorganisms by participating in one or more chemical reactions that damage cell components –
   - Soaps and detergents remove microbes from oily or greasy surface.
   - Lactic and propionic acids retard mold growth in breads and other bakery products.
   - Heavy metals like selenium, mercury, copper and silver can be very effective in inhibiting bacterial growth.
   - Hypochlorous acid, formed by the addition of chlorine to water, effectively controls microorganisms in drinking water and swimming pools.
   - Ethyl and isopropyl alcohols can be used as skin antiseptics.
   - Phenol derivatives can be used on skin, instruments, utensils and furniture. It is also used to destroy discarded cultures.
2. **Physical agents:**
   a. Heat: It is a preferred agent of sterilization for all materials not damaged by it.
      - Dry heat does most of its damage by oxidising molecules. It is usually used to sterilize metal objects and glassware and is the only suitable means of sterilizing oils and powders.
      - Moist heat, because of its penetrating properties, is a widely used physical agent. The autoclave, which uses moist heat under pressure, is a common instrument for sterilization.
      - Pasteurization kills most pathogens in milk, other dairy products and beer but does not sterilise them. Milk is pasteurized by heating it to 71.6^{o}C for at least 15 seconds in flash method or by heating it to 62.9^{o}C for 30 minutes in the holding method.

   b. Low temperature: Refrigeration, freezing, freeze-drying can be used to retard the growth of microorganisms by slowing the rate of enzyme-controlled reactions but does not kill the microbes.

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c. Radiation: Four general types of radiation – ultraviolet light, ionizing radiations, microwave radiations and strong visible light (under certain circumstances) – can be used to control microorganisms and to preserve food.
d. Filtration: Generally this method is opted to sterilize substances that are destroyed by heat, like, hormones, enzymes etc.

Media compositions:

Growing bacteria in the laboratory requires knowledge of their nutritional needs and the ability to provide the needed substances in a medium.
The culture medium consists of chemicals which support the growth of culture or microorganisms. Microbes can use the nutrients of culture media as their food which is thus necessary for cultivating them *in vitro*.
A good culture medium contains sources of carbon, nitrogen, hydrogen, oxygen, phosphorus, inorganic salts and other growth promoting substances. Constituents of basal culture media include:

- Water: Source of hydrogen and oxygen.
- Electrolyte: Source of inorganic salts.
- Glucose: Source of carbon and energy.
- Peptone: It is a complex mixture of partially digested protein. Source of nitrogen.
- Agar: Solidifying agent of culture medium.

Nutrient agar medium is a general purpose medium supporting growth of a wide range of bacterial species. It typically contains (mass/volume) –

- Peptone (0.5%): This provides organic nitrogen.
- Beef extract / Yeast extract (0.3%): The water-soluble content of these contribute vitamins, carbohydrates, nitrogen and salts.
- Agar (1.5%): This gives the mixture solidity.
- Sodium Chloride (0.5%): This gives the mixture proportions similar to those found in the cytoplasm of most organisms.
- Distilled water: It serves as a solvent and transport medium for the medium's various substances.

These ingredients are added, stirred, pH adjusted to 7 at 25°C and boiled for approximately one minute to ensure they are mixed and then sterilized by autoclaving, typically at 121.5^{o}C for 15 minutes. After autoclaving, the medium is cooled to around 50°C and poured into Petri dishes, which are covered immediately. Once the dishes hold solidified agar, they are ready to use for culturing microorganisms. They can be used immediately or stored upside down and are refrigerated.

Types of media:

1. On the basis of chemical composition of medium:
    - Complex medium / chemically non-defined medium: Such medium contains reasonably familiar materials but varies slightly in chemical composition from batch to batch.
    - Defined synthetic medium: Such medium contains exactly known specific kinds and amounts of chemical substances.
2. On the basis of isolation and identification of particular microorganisms:
    - Selective medium: Such medium encourages the growth of some microorganisms, but supresses the growth of others.

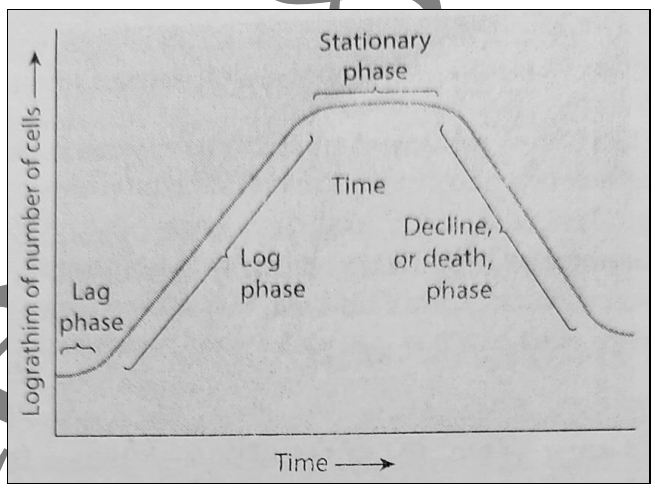
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- Differential medium: Such medium has constituent that causes an observable change (change in colour or pH) in the medium when a particular biochemical reaction occurs. This change allows microbiologists to distinguish a certain type of colony from others growing on the same plate.
- Enrichment medium: Such medium contains special nutrients that allow growth of a particular type of organism that might not otherwise be present in sufficient numbers to allow it to be isolated and identified. Unlike a selective medium, an enrichment medium does not supress the growth of others.

Growth kinetics:

Growth is the fundamental, dynamic biological feature exhibited by the populations of all species. If more number of individuals is added to the population, the population shows a positive growth. It will show a negative growth if a number of individuals are removed or lost. The population will shoe zero growth if the number of individuals added or removed is the same.
Microbial growth is defined not in terms of cell size but as the increase in the number of cells, which occurs by cell division. Cell division in bacteria, unlike cell division in eukaryotes, usually occurs by binary fission or sometimes by budding.

Phases of growth:
Considering a population of microorganisms introduced into a fresh, nutrient-rich medium, a mixture of substances on or in which microorganisms grow. Such microorganisms display four major phases of growth –



1. Lag phase: In the lag phase, the organisms do not increase significantly in number, but they are metabolically active, growing in size, synthesizing enzymes and incorporating various molecules from the medium.
2. Log phase: Once organisms have adapted to the medium, population growth occurs at an exponential or logarithmic rate. During the log phase, the organisms divide at their most rapid rate.
3. Stationary phase: When cell division decreases to the point that new cells are produced at the same rate as old cells die, the number of live cells stays constant.
4. Decline (death) phase: As conditions in the medium become less and less supportive of cell division, many cells lose their ability to divide, and thus the cells die. In this decline phase, or death phase, the number of live cells decreases at a logarithmic rate.

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